

Git Assignment: Collaboration & Branch Management

Objective

Students will learn to manage branches, simulate collaboration, handle merge conflicts, and work with remote repositories.

Part 1: Repository Setup

1. Create project folder

```
mkdir mid-git-project  
cd mid-git-project
```

2. Initialize Git

```
git init
```

3. Create files

```
echo "# Mid Level Git Project" > README.md  
echo "console.log('App started');" > app.js
```

4. Commit

```
git add .  
git commit -m "Initial project setup"
```

Part 2: Branching Strategy

Create branches:

```
git checkout -b develop  
git checkout -b feature-login
```

Part 3: Feature Development

Login Feature:

Modify app.js and commit:

```
git add .  
git commit -m "Added login feature"
```

Dashboard Feature:

```
git checkout develop
```

```
git checkout -b feature-dashboard  
Add code and commit.
```

Part 4: Merging

```
git checkout develop  
git merge feature-login  
git merge feature-dashboard
```

Part 5: Merge Conflict

Edit same line in both branches and merge.
Resolve conflict manually, then:
git add app.js
git commit -m "Resolved merge conflict"

Part 6: Remote Repository

```
git remote add origin https://github.com/your-username/mid-git-project.git  
git push -u origin main  
git push origin develop
```

Submission Requirements

Submit GitHub link and screenshots of branches, conflicts, and commit history.

<https://forms.gle/YbxvwDz8FaH5AX718>